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Underwater laser communication is a technique that employs laser propagation under water to transmit data from one point to another. The technology is useful where physical connections through fibre optic cables are impractical due to high costs or other considerations.

Free Space Optics (FSO) technology is based on connectivity between FSO-based optical wireless terminals, each consisting of an optical transceiver to provide full duplex (bi-directional) communication functionality. Each optical wireless terminal uses an optical source (laser LED) and a telescope that transmits light through the atmosphere to the receiver. The receiver consists of an optical system and an opto-electronic front end that, in turn, comprises a pin photodiode or avalanche photodiodes, which convert the optical input into an electrical output.

FSO technology for high-speed and secure data transmission was originally developed by the US military and NASA. It has been used for more than three decades in various forms to provide fast communication links in remote locations. Over the years, FSO technology has gained acceptance in the telecommunication industry, particularly in the enterprise campus networking environment. Due to the enormous bandwidth capabilities of FSO transmission and the worldwide unlicensed nature of the transmission spectrum, FSO technology has enormous potential for short-distance wireless connections.

Apart from the obvious advantages of the high rates of laser communication underwater, such transmission is also inexpensive, immune to jamming and has a low signal-to-noise ratio. However, line-of-sight is a major disadvantage. The basic concept of laser communication between two computers, both in free space and under water, is more or less the same. However, factors that impact performance underwater, like water salinity and turbidity,

Underwater and Other Emerging Trends in Laser Communication

Underwater laser communication poses a number of challenges today. However, this technology is being researched extensively as it has a wide range of military and other applications



have to be taken into consideration.

In this article, design of the actual concept of underwater communication is proposed to be configured around green laser module that is intensity modulated by the intelligence signal to be transmitted. On the receiver side, a suitable photo-sensor-based circuit is used to demodulate the signal. On the transmitter and receiver side, the computer interface is through RS-232. The data file to be transmitted is converted into a serial bit stream, and then used to modulate the laser through the RS-232 serial interface. On the receiver side, the laser beam is detected and the serial bit stream is recovered. The bit stream is converted back to the desired format in the PC. The latter part of the article cov-

ers the advantages of laser communication over optical communication under water, and also discusses the emerging concepts and their implementation.

Operational methodology

The following section covers the operational principle of underwater laser communication links. The proposed communication setup comprises five major elements—the source of the data to be transmitted (in this case, the PC), the transmitter, communication channel, receiver and the data receiving terminal (PC).

Transmitter

The data file in the PC, which is to be transmitted to another remote PC, is first